MA1522 Linear Algebra for Computing

Instructions for Final Examination

Semester 1, 2026/27

Final Exam Overview

- Date and Time: 24 November 2026 (Tuesday), 17:00–19:00.
  Duration: 2 hours.

- Venue: MPSH2A and MPSH2B. Check your seat number in **myEduRec**.

- Exam Format: Face-to-face, **closed book with one A4-sized helpsheet**. Helpsheet can be handwritten or typed on both sides.

- Calculator Policy: Only calculators in NUS list of approved calculators are permitted. This list can be found in Canvas – Files.
  Show full working for all computations.

- Things to bring: Student card, dark pencil, approved calculator, A4-sized helpsheet.

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Exam Structure

- Paper consists of 2 sections, Section A and Section B, for a total of 100 marks.
- Section A: 16 multiple-choice questions, 2 marks each. Shade your answers for Section A using dark pencil on the OMR (Optical Mark Recognition) sheet provided.
- Section B: 5 written questions, 68 marks total. Marks allocated to each question may not be the same, and are indicated clearly at the beginning of the question. Write your answers directly in the answer book provided.
- Indicate your **student number** in the designated fields on the OMR sheet and the answer booklet. Do not write your name.

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Question Paper Cover Page

NATIONAL UNIVERSITY OF SINGAPORE

MA1522 LINEAR ALGEBRA FOR COMPUTING

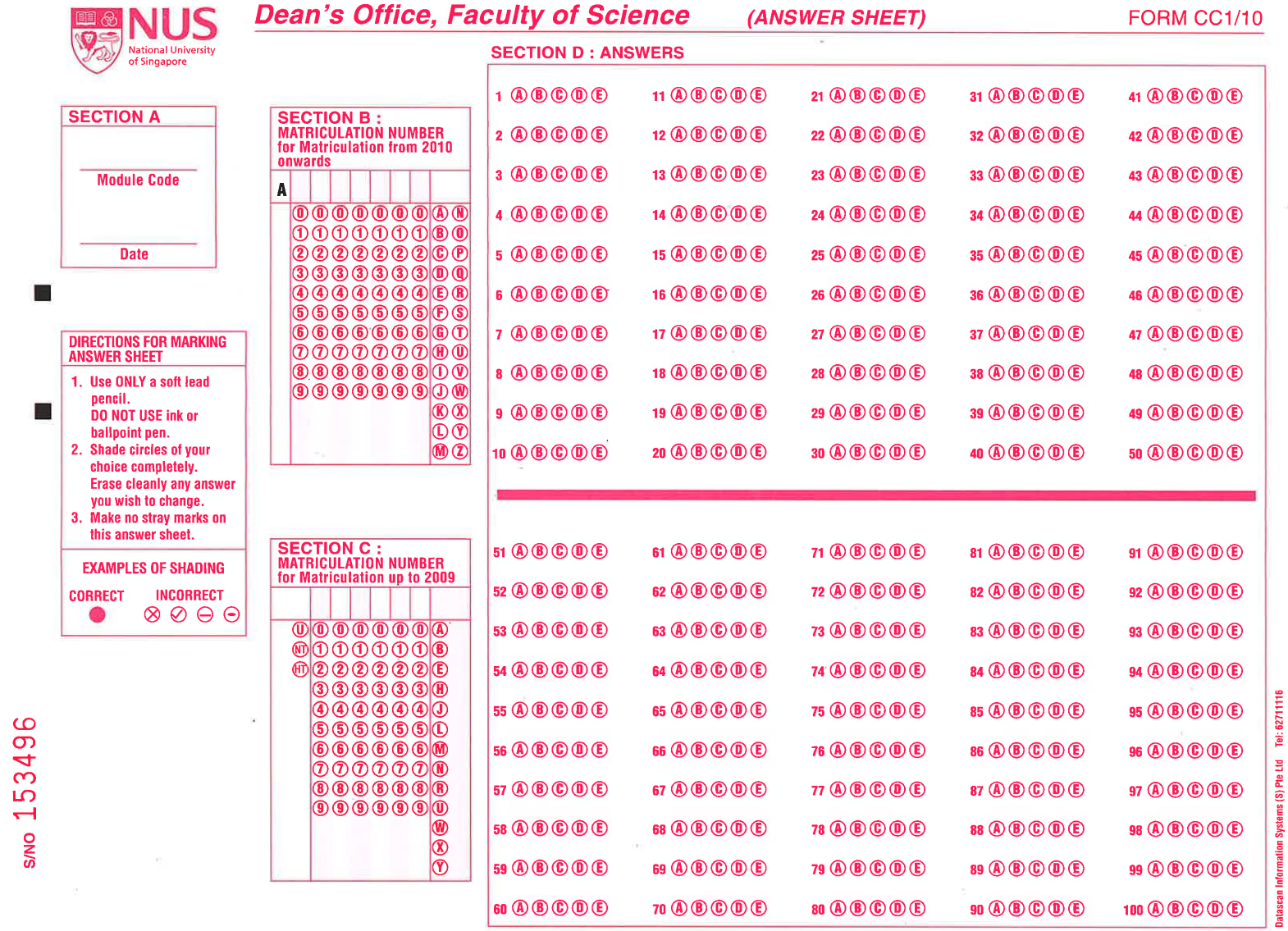
(Semester 1 : AY 2026–2027)

Time allowed : 2 hours

INSTRUCTIONS TO CANDIDATES

1. This question booklet consists of **TWO (2)** sections and comprises **8** printed pages, including this cover page.
2. Section A contains **SIXTEEN (16)** multiple-choice questions, and each question carries **TWO (2)** marks.
3. Section B contains **FIVE (5)** questions. The marks for each question may not be the same, and are clearly indicated at the beginning of the question.
4. Answer **ALL** questions in both sections.
   (a) Write your answers for Section A on the **OMR sheet** provided.
   (b) Write your answers for Section B in the **answer book** provided.
5. This examination carries a total of **ONE HUNDRED (100)** marks.
6. This is a **CLOSED BOOK** examination with **ONE (1) A4-sized helpsheet**.
7. You may use graphing calculators from the NUS approved list. However, you should lay out systematically the various steps in the calculations.

OMR Sheet



NUS National University of Singapore

Dean's Office, Faculty of Science *(ANSWER SHEET)* FORM CC1/10

SECTION A

Module Code

Date

DIRECTIONS FOR MARKING ANSWER SHEET

1. Use ONLY a soft lead pencil. DO NOT USE ink or ballpoint pen.
2. Shade circles of your choice completely. Erase cleanly any answer you wish to change.
3. Make no stray marks on this answer sheet.

EXAMPLES OF SHADING

CORRECT ● INCORRECT ⊗ ⊘ ⊖ ⊝

SECTION B : MATRICULATION NUMBER for Matriculation from 2010 onwards

A

SECTION C : MATRICULATION NUMBER for Matriculation up to 2009

U NT HT

SECTION D : ANSWERS

1 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	11 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	21 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	31 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	41 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
2 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	12 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	22 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	32 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	42 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
3 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	13 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	23 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	33 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	43 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
4 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	14 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	24 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	34 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	44 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
5 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	15 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	25 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	35 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	45 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
6 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	16 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	26 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	36 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	46 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
7 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	17 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	27 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	37 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	47 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
8 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	18 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	28 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	38 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	48 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
9 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	19 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	29 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	39 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	49 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
10 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	20 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	30 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	40 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	50 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
51 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	61 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	71 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	81 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	91 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
52 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	62 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	72 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	82 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	92 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
53 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	63 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	73 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	83 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	93 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
54 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	64 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	74 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	84 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	94 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
55 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	65 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	75 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	85 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	95 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
56 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	66 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	76 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	86 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	96 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
57 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	67 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	77 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	87 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	97 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
58 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	68 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	78 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	88 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	98 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
59 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	69 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	79 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	89 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	99 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ
60 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	70 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	80 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	90 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ	100 Ⓐ Ⓑ Ⓒ Ⓓ Ⓔ

S/NO 153496

Datascan Information Systems (S) Pte Ltd Tel: 62711116

Answer Book Cover Page

National University of Singapore

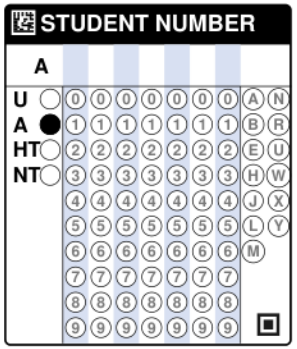
MA1522 – LINEAR ALGEBRA FOR COMPUTING

ANSWER BOOK FOR SECTION B

AY2026/2027 Semester 1

INSTRUCTIONS TO STUDENTS

1. Write your **student number** in the grid on the right and shade the corresponding circle **completely** for each digit or letter, using ink or dark pencil. **DO NOT WRITE YOUR NAME.**
2. This section contains **FIVE (5)** questions. Answer **ALL** questions.
3. Write your answer to each part **only in the designated box**. Each part has a single answer box, even when two parts share a page.
4. If you need more space, use the **additional pages at the end**. On the original page, **tick the box** at the bottom and write the page number you continued on. On the additional page, state the question and part number.
5. This booklet comprises **16 printed pages**, including this cover page and two additional pages at the end.
6. This is a **CLOSED BOOK** examination with **ONE (1) A4-sized helpsheet**.
7. You may use graphing calculators from the NUS approved list. However, you should lay out systematically the various steps in the calculations.
8. Hand in this entire booklet at the end of the examination.



Answer Book for Section B

- Write your answers for each question in the **allocated pages** of the answer booklet. Write neatly within the space provided.
- If you run out of space, continue your answer on the additional pages at the back of the answer book. Clearly indicate:

  On the original page: the page number where your answer continues.

  On the additional page: the question number and part number, and the page from which your answer continues.

Answer Book for Section B

- Write your answers for each question in the **allocated pages** of the answer booklet. Write neatly within the space provided.

- If you run out of space, continue your answer on the additional pages at the back of the answer book. Clearly indicate:

  On the original page: the page number where your answer continues.

  On the additional page: the question number and part number, and the page from which your answer continues.

Question 1(b): Determine the matrix $\boldsymbol{A}$. (4 marks)

☐ I have continued this answer on additional page(s) number: _____

Additional working space

Use this page only if you ran out of space. Tick the box on the original answer page and state clearly which part you are continuing here.

Question: _____ **Part:** _____ **Continued from page:** _____

Before the Exam

- Confirm your exam venue and seat number in advance. Pay close attention to the venue!
- Arrive at the exam venue by 16:45. Entry starts at 16:50.
- Bring your **student card or valid physical ID**.
- Scan the QR code at the exam venue entrance for attendance.
- Remember to bring a dark pencil for shading the OMR sheet and your student number on the Section B answer book, as well as an approved calculator.
  Invigilators will not have these items available for loan.

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During the Exam

- Students are not allowed to leave during the first hour or the last 15 minutes of the exam.
- Students are not allowed to enter the exam venue after one hour from the start time.
- Put your **mobile phone and all smartwatches** in your bag and place your bag at the front/side/back of the exam hall.
- Ensure that all alarms, notifications, and timers are turned off before the exam begins. An alarm going off during the exam is extremely disruptive and may constitute a breach of examination regulations.
- Mobile phones or communication devices must not be brought along when going to the washroom.

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Academic Integrity

- All students must uphold the highest standards of **academic integrity** during the exam.
- Obey all instructions, including to stop writing when told to do so.
- Any breach of academic integrity will be dealt with in accordance with **university regulations**.

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Final Reminder

Remember to bring:

- Student card
- Dark pencil
- Approved calculator
- Helpsheet

Questions

All queries should be posted in Canvas Discussion - Final Exam.

Queries by email will be redirected there.